$$u^* \nabla\cdot(\nabla u) - u \nabla\cdot(\nabla u^*) \quad ①$$

↓

$$\nabla\cdot(u\nabla u^* - u^*\nabla u) \quad ②$$

$$② = \nabla\cdot u \nabla u^* + u\nabla^2 u^* - \nabla\cdot u^* \nabla u - u^*\nabla^2 u$$

$$= u\nabla^2 u^* - u^*\nabla^2 u$$

$$u = \sqrt{I(x)}\, e^{ikx}$$

$$\sqrt{I(x)}\, e^{-ikx}$$

$$\nabla\cdot u \nabla u^* - \nabla\cdot u^* \nabla u = 0 \ ?$$

~~$u\cdot\nabla^2 u^*$~~

$$u\nabla u^* - u\nabla^2 u^* - u^*\nabla u + u^*\nabla^2 u$$

$$(u\nabla u^*)^* - u^*\nabla u$$

$$\left(\frac{u^*}{u}\right)'$$

$$\left(\frac{u u^*}{u^2}\right)'$$

$$\nabla U = \sqrt{I}\exp(ikx)$$

$$ik\sqrt{I}\exp(i\phi(x)) + \frac{d\sqrt{I}}{dx}\exp(ikx)$$

$$ikU\nabla\phi$$